UNIVERSITI TUNKU ABDUL RAHMAN
LEE KONG CHIAN FACULTY OF ENGINEERING AND SCIENCE
LABORATORY SAFETY RULES

The following rules must be obeyed by all students in the Computer laboratories of the faculty. Willful or repeated inadvertent noncompliance may result in dismissal or suspension from the laboratories.

I. No entry without permission:

i) Outsiders are not allowed to enter the laboratory without permission.

ii) No student is allowed to enter the laboratory unless permission has been given by a laboratory staff or a lecturer.

II. At work in the laboratory:

i) The computer facilities at the laboratory are to be used for learning purposes only.

ii) Wear your I.D. cards at all times in a proper visible position whenever you come into the computer laboratory.

iii) It is compulsory to follow dress code of UTAR
   a) Covered shoes (no slippers or flip flops)
   b) Long pants and tidy attire

iv) Do not eat or drink in the laboratory.

v) Do not tamper with any computers, printers, or any other associated equipment. If there are problems or faults, report immediately to the staff at the laboratory.

vi) Do not remove any equipment or document from the computer laboratories.

vii) Do not alter the default hardware and software setup of the computers and printers in the computer laboratories.

viii) Do not copy, rename, change, examine or delete files or information belonging to any other user of the University.

ix) Do not install any unauthorized or unlicensed software into the PC of computer laboratories.

x) Do not make any noise, chit-chat, hold group discussions or disturb other users at the laboratory in any way.

xi) Silent your hand phone when you are in the computer laboratories.

xii) Any equipment broken or damaged must be reported immediately to the laboratory staff.

III. Before leaving the laboratory:

i) No student is allowed to take away any equipment, tools or other property of the laboratory without permission.

ii) Please check and remember to take all your personal belongings such as handphone, wallet, pendrive, hard disk, book, bottle, sweeter, and etc.

iii) Lab staffs are not responsible for lost of personal belongings.

Universiti Tunku Abdul Rahman

Faculty	**Lee Kong Chian Faculty of Engineering and Science**
Department:	**Department of Electrical and Electronics Engineering**
Unit Code and Name	**UEEN 1043 Object-Oriented Concepts and Programming Techniques**
Experiment No.:	**4 / 7**
Title of Experiment:	**Advanced Data Types**
Laboratory Room No. and Name:	**Computer lab**
Experiment Duration (hour):	**3 hours**
Number of Student per Group	**1 student**

Objectives

- To create practical Python program using container datatypes.

Equipment and Materials

Item Description	***Item category**	Quantity estimation (e.g. per set/group of student)
Personal computer	E	1 per student
Python interpreter (Anaconda Spyder)	S	1 per student

***Item category**	
SP	Sample or specimen
C	Consumable
CH	Chemical
W	Labware, glassware, tool, and components
E	Equipment
S	Software

Caution Statement (Optional)

FACULTY OF ENGINEERING AND SCIENCE
UNIVERSITI TUNKU ABDUL RAHMAN

UEEN 1043 Object-Oriented Concepts and Programming Techniques

Practical 4: Advanced Data Types

Objectives

- To create practical Python program using container datatypes.

Equipment

- Personal computer with a Python interpreter (Anaconda Spyder)

Introduction

Launch Anaconda Spyder IDE for Python and write codes for the following.

Exercise

4-1 Dictionary

Dictionary stores pairs of items, called key-value pair.

```
price = {}     #an empty dictionary
price = { 'orange': 0.80, 'apple': 0.50, 'pear': 1.20 }
               #initialize a dictionary
price.keys()   #get all keys of dict
min( price.keys() )
max( price.keys() )
price.values() #get all values of dict
price['banana'] = 0.20
               #add a new entry, or update an existing entry
price.update( { 'durian': 200 } )  #update an existing entry
del price['orange']                #remove and entry
if 'orange' in price:              #check for an entry
    print( price['orange'] )
if 0.5 in price.values():
    print('selling cheap' )

for x in sorted( price.keys() ):
     print(x)  #str

for y in price.values():
     print(y)  #float
```

4-2 Practice creating dictionary

Assume a list **B**, where

```
B = [ 'Alan', 'Age', 23, 'Gender', 'Male',
      'Bobby', 'Age', 28, 'Gender', 'Male',
      'Catherine', 'Age', 21, 'Gender', 'Female' ]
```

Write a function to convert the list above into a dictionary **A**, such that the statement 'print (A)' will produce the answer shown below.

```
{'Alan': {'Age': 23, 'Gender': 'Male'},
'Bobby': {'Age': 28, 'Gender': 'Male'},
'Catherine': {'Age': 21, 'Gender': 'Female'}}
```

4-3 Manipulating dictionary

A dictionary consists of students' data is given as follows:

```
students = {'id1':
   {'name': ['Sarah'],
    'class': ['A'],
    'courses': ['english, math, science']
   },
 'id2':
  {'name': ['Ivan'],
    'class': ['B'],
    'courses': ['english, math, science']
   },
 'id3':
    {'name': ['Sarah'],
    'class': ['A'],
    'courses': ['english, math, science']
   },
 'id4':
   {'name': ['Lee'],
    'class': ['A'],
    'courses': ['english, math, science']
   },
  'id5':
   {'name': ['Zoe'],
    'class': ['B'],
    'courses': ['english, math, science']
   },
}
```

1. Remove duplicates from the dictionary. The expected content of the dictionary should be:

```
{'id1': {'name': ['Sarah'],
  'class': ['A'],
  'courses': ['english, math, science']},
 'id2': {'name': ['Ivan'],
  'class': ['B'],
  'courses': ['english, math, science']},
 'id4': {'name': ['Lee'],
  'class': ['A'],
  'courses': ['english, math, science']},
 'id5': {'name': ['Zoe'],
  'class': ['B'],
  'courses': ['english, math, science']}}
```

2. Rearrange the dictionary, sorted by the students' name. The expected content of the dictionary should be:

```
{'id2': {'name': ['Ivan'],
  'class': ['B'],
  'courses': ['english, math, science']},
 'id4': {'name': ['Lee'],
  'class': ['A'],
  'courses': ['english, math, science']},
 'id1': {'name': ['Sarah'],
  'class': ['A'],
  'courses': ['english, math, science']},
 'id5': {'name': ['Zoe'],
  'class': ['B'],
  'courses': ['english, math, science']}}
```

3. Update the dictionary with new scores data

```
scores = {'id1': 70, 'id2': 75, 'id3': 70, 'id4': 80, 'id5': 60}
```